



VIT[®]
BHOPAL

Project Report

Name	Priyansh Vivek
Reg. No.	25BAI10834
Branch	CSE (AI&ML)
Year	First
Course Code	CSA2001
Date	29 March 2026

AI-Driven Medicine Recommender

The Problem :-

Access to accurate and quick medical guidance for minor ailments is often a challenge, especially in areas with limited access to pharmacies or healthcare professionals. Many people are unsure which over-the-counter (OTC) medications are appropriate for common symptoms like a cough, fever, or an upset stomach. Without guidance, individuals may choose the wrong medication, experience delays in relief, or unnecessarily visit a clinic for minor conditions.

This project addresses that gap by providing a fast, locally-run AI-powered assistant that recommends appropriate OTC medications based on described symptoms.

Why It Matters :-

In a world where information is abundant but often disorganized, creating a streamlined, **accessible tool for self-care** is vital. Many people struggle to identify which Over-the-Counter (OTC) medication is appropriate for common, non-emergency symptoms.

This project matters because:

- **Accessibility:** It provides instant, categorized guidance for minor ailments.
- **Safety Education:** By linking symptoms to specific active ingredients (like *Guaifenesin* for wet coughs), it helps users understand *how* medications work rather than just memorizing brand names.
- **Efficiency:** It reduces the "search fatigue" of browsing endless medical forums by providing a direct, dictionary-backed response.

Approach to Solving the Problem :-

The goal was to build a "low-latency, high-accuracy" assistant that didn't require a massive neural network to function.

- **Knowledge Engineering:** I structured the data as a Python Dictionary. This allows for (constant time) lookup speed once a symptom is identified.
- **Natural Language Processing (NLP) Lite:** Instead of a complex AI model, I used difflib , which employs the Ratcliff/Obershelp algorithm to compute the similarity between two strings. This allows the program to handle typos and variations.
- **User Interface:** I implemented a Command Line Interface (CLI) loop to ensure a continuous conversation until the user explicitly chooses to exit.

Key Decisions :-

- **Fuzzy Matching vs. Exact Matching:** I chose a cutoff=0.4 in the get_close_matches function. This was a deliberate balance - too high, and it wouldn't catch typos; too low, and it might suggest "Fever" when the user types "Flu."
- **Categorization:** I grouped the knowledge base into five distinct domains (Respiratory, Digestive, Skin, Eye/Ear, Pain) to help the user understand the scope of the AI's "brain."
- **Title Case Normalization:** I used .title() on user input to ensure that "fever," "FEVER," and "Fever" all map correctly to the dictionary keys.

Challenges Faced :-

- **Ambiguity in Symptoms:** One major challenge was handling symptoms that overlap. For example, "Diarrhea" and "Loose Motion" lead to the same treatment. I resolved this by adding redundant keys in the dictionary to catch different ways people describe the same problem.
- **The "No Match" Scenario:** Early versions of the code crashed if a match wasn't found. I had to implement error handling logic in the find_match method to return None gracefully.

- **Formatting for Readability:** Printing raw dictionary data is messy. I decided to use f-strings with Emojis (🔍, 💊, 📄) to make the output "scannable" for a user who might be feeling unwell and frustrated.

What I Learned :-

- **Data Structure Design:** I learned that the way you structure your "Knowledge Base" is just as important as the code that searches it. A clean dictionary makes the logic much simpler.
- **Algorithmic Trade-offs:** I gained experience in tuning "sensitivity." I realized that in a medical context, it is better to say "I don't know" than to give a high-confidence wrong answer.
- **The Importance of Disclaimers:** Building this project highlighted the **ethical responsibility** of AI developers. Even a simple script must include a disclaimer to ensure it isn't mistaken for professional medical diagnosis.